

PLAYHACK · SDE TRACK · IIT GUWAHATI

PLAYHACK

Campus sports facility booking where the winner of a race for one court is decided by Postgres, not by our code.

TEAM **INNOVAIT**

INNOVAIT-HACKATHON.VERCEL.APP

0 OVERLAPPING BOOKINGS, EVER

THE PROBLEM, IN THE BRIEF'S OWN WORDS

AT 6 PM, FIFTY STUDENTS WANT THE SAME COURT

"The system must confirm **exactly one** and reject the rest, without corrupting data."

Almost every booking system answers this the same way — and almost every one of them is wrong.

```
// the question
const taken = await isSlotTaken(court, slot);

// the answer
if (!taken) await insertBooking(court, slot);
```

Both lines are individually correct. Between them is a **gap** — and under load another request commits inside it.

THE ANSWER

THE WRITE *IS* THE DECISION

```
ALTER TABLE bookings ADD CONSTRAINT bookings_no_overlap  
    EXCLUDE USING gist (facility_id WITH =, during WITH &&)  
WHERE (status = 'confirmed');
```

There is no question and no answer. A booking carries a time range, and the database **physically cannot store two overlapping ranges** for the same facility. The loser is rejected with SQLSTATE **23P01** before it ever becomes a row — no availability check on the write path, and no application lock on the slot.

PARTIAL OVERLAP IS THE CASE THAT HAPPENS

	UNIQUE (FACILITY, STARTS_AT)	EXCLUDE ... WITH &&
Two bookings at 18:00	rejected	rejected
18:00–19:00 vs 18:30–19:30	both stored	rejected
A two-hour closure over four slots	not expressible	one row
Cancelling frees the slot	needs a delete	status flip

A booking from 18:30 does not share a start time with one from 18:00 — but it **overlaps** it, and overlap is what matters to a student holding a racket.

THE GAP BETWEEN THE QUESTION AND THE ANSWER

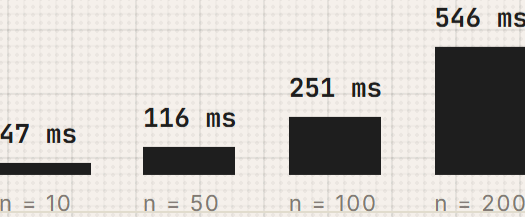
Fifty students tap Book on the same court at 18:00. The same burst, fired twice at the deployed app — once at a textbook availability check, once at a database that refuses to store the overlap.

1225

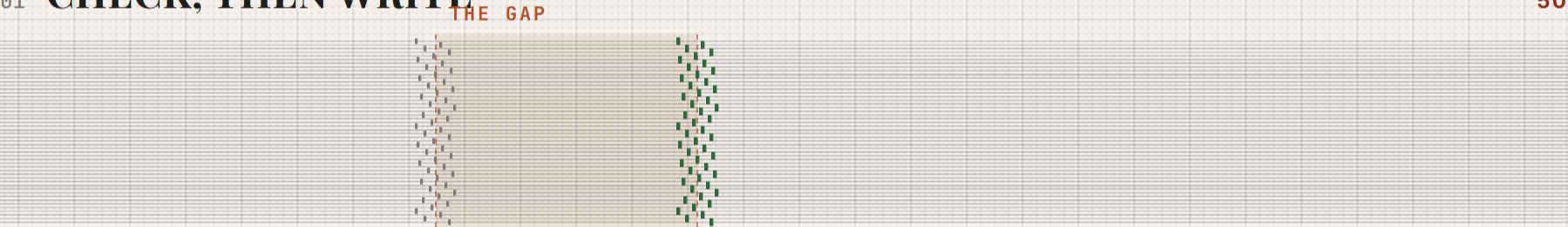
overlapping pairs left behind by **check-then-write**, swept across the whole table

0

from the identical burst, once the constraint is carrying the decision



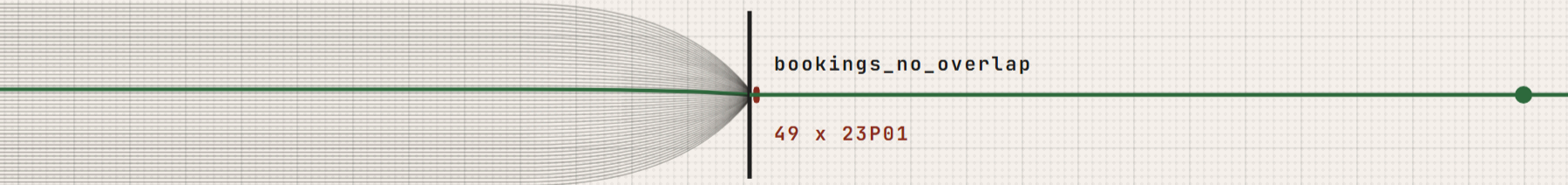
01 CHECK, THEN WRITE



50 CONFIRMED · NOTHING ERRORED

Every request reads “free”, then writes. The shaded interval is the gap between the question and the answer — and fifty answers commit inside it.

02 THE WRITE IS THE DECISION



1 CONFIRMED · 49 REJECTED

No read. Fifty inserts race one exclusion constraint; Postgres rejects forty-nine with `SQLSTATE 23P01` before they can become rows.

```
ALTER TABLE bookings
ADD CONSTRAINT bookings_no_overlap
EXCLUDE USING gist (
  facility_id WITH =,
  during WITH &&
) WHERE (status = 'confirmed');
```

Measured on the deployed app — Vercel **sin1**, Neon Postgres **ap-southeast-1**, warm, median of three. Server time is **sub-linear in n**: twenty times the load for ten times the time. The overlap count is a whole-table sweep after every run, not a check of the slot just tested.

THE SAME BURST, FIRED TWICE AT PRODUCTION

FIFTY REQUESTS. ONE COURT.

CHECK, THEN WRITE

50 bookings confirmed. Nothing errored.

1225 overlapping pairs, swept across the whole table

Fifty students walk to the same court.

THE WRITE IS THE DECISION

1 confirmed — 49 rejected with 23P01

0 overlapping pairs, across 380+ confirmed rows

Every rejection is typed and carries three alternative slots.

MEASURED ON THE DEPLOYED APP

SUB-LINEAR, NOT LINEAR

All contenders attempt the insert at once. One wins; the rest block on its uncommitted row and are released together the moment it commits. **Nothing in the write path serialises on n.**

Vercel **sin1**, Neon **ap-southeast-1**. Median of three warm runs.

MODE	REQUESTS	SERVER	CONFIRMED	OVERLAPS
naive	50	37 ms	50	1225
safe	10	47 ms	1	0
safe	50	116 ms	1	0
safe	100	251 ms	1	0
safe	200	546 ms	1	0

CORRECTNESS IS THE FLOOR

Idempotency keys. A retried submit returns the original booking. A double-tap on hostel wifi is the common case, not the edge case.

Typed rejections. Losing returns `SLOT_TAKEN`, `QUOTA_EXCEEDED`, `OVERLAPS_OWN` — with three alternative slots and a queue position.

Waitlist promotion. Cancelling frees the slot *and* offers it to the next student in the same transaction — no window where it is free but unowned.

Maintenance closures. A closure is a booking row. One invariant, two features: a manager cannot close a court out from under a confirmed reservation.

Fair draw. Peak slots go to a seeded, weighted, published lottery. First-come-first-served rewards the best wifi, not the keenest player.

THE PRODUCT

BUILT FOR A STUDENT ON A PHONE AT 5:58 PM

Indian Institute of Technology Guwahati | Students' Sports Board

12:14 IST

PLAYHACK

CAMPUS FACILITY BOOKING

BOOKMY BOOKINGSRACEFAIR DRAWINSIGHTSOPS

D

READING AS

Demo Student

SWITCH READER

← BACK TO THE GRID

TENNIS

TENNIS COURT A

SPORTS COMPLEX, NEAR LOHIT

Floodlit hard court. Evening slots go fast.

HOURS

06:00-21:00

SLOT

60 min

CAPACITY

4

PLAYING SURFACE, MARKINGS TO SCALE

01

SUN 30 AUG

13 OF 15 OPEN

A slot is yours only once the confirmation appears. Nothing is held by clicking.

TODAY

TOMORROW

MON 31 AUG

TUE 1 SEPT

WED 2 SEPT

THU 3 SEPT

FRI 4 SEPT

SAT 5 SEPT

OPEN

YOURS

TAKEN

CLOSED

06:00

OPEN

07:00

OPEN

08:00

OPEN

09:00

OPEN

10:00

OPEN

11:00

OPEN

12:00

OPEN

13:00

OPEN

14:00

PRISHA GOGOI

15:00

OPEN

16:00

OPEN

17:00

OPEN

18:00

PEAK

19:00

PEAK

20:00

PEAK

LIVE

Booking.

The day as a fixture table, live over SSE.

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12:14 IST

PLAYHACK

CAMPUS FACILITY BOOKING

BOOKMY BOOKINGSRACEFAIR DRAWINSIGHTSOPS

D

READING AS

Demo Student

SWITCH READER

INNOVATION · FAIR ALLOCATION

WINNING A RACE IS NOT DESERVING THE COURT

The exclusion constraint guarantees that exactly one booking survives a stampede. It does not decide who should win — and first-come-first-served decides that badly. At six the winner is whoever has the best wifi, the newest phone, or a script.

So peak slots are not released as a race. Requests arriving inside a short window become entries, and one weighted, seeded draw picks the winner. Fairness sits on top of correctness: the winner's booking still goes through the same constrained path as any other.

01

HOW THE DRAW WORKS

A draw a student cannot explain to the person who lost is not fair, only opaque.

01

A WINDOW, NOT A STARTING PISTOL

Every request that arrives inside the window becomes an entry. Arriving in the first millisecond buys nothing, so there is no advantage in a faster phone, a closer router, or a script left running.

02

WEIGHT BY RELIABILITY

Each entry carries weight 50 + reliability/2. A student who turns up keeps a full ticket; a record of no-shows costs some of it. The floor of 50 means nobody is ever locked out.

03

SEEDED, AND PUBLISHED

The winner is sha256(seed || entries) over the entrant list sorted by id. The seed is published with the result, so anyone can recompute the winner and check it.

02

RUN A DRAW

Opening a window clears the slot first, so every draw starts from the same state.

Fair draw. Rules and seed published before the draw.

Indian Institute of Technology Guwahati | Students' Sports Board

12:14 IST

PLAYHACK

CAMPUS FACILITY BOOKING

BOOKMY BOOKINGSRACEFAIR DRAWINSIGHTSOPS

D

READING AS

Demo Student

SWITCH READER

OPERATIONS · INSIGHTS

WHERE THE CAMPUS ACTUALLY PLAYS

Every figure on this page aggregates the same bookings table the booking path writes to. There is no reporting copy, no nightly export and nothing that can drift out of step with what actually happened.

UPCOMING BOOKINGS

152

BOOKED LAST 7 DAYS

423

ACTIVE STUDENTS

199

ON WAITLISTS

12

NO-SHOW RATE

13%

01

DEMAND BY HOUR

Bookings started in each hour, trailing 14 days. Solid bars are the peak band.

3

18

79

73

75

76

74

82

90

79

96

94

100

99

32

33

21

Mon

Tue

Wed

Thu

Fri

Sat

Sun

Mon

Tue

Wed

Thu

Fri

Sat

Sun

Mon

Tue

Wed

02

UTILISATION BY FACILITY AND HOUR

Denser hatching means busier. An empty band at the right of a row is capacity nobody is using.

Insights. Aggregated from the booking table itself.

TWELVE FACILITIES · AVAILABILITY DERIVED, NEVER STORED

09

VERIFIED, NOT ASSERTED

EVERY CLAIM ON THESE SLIDES IS REPRODUCIBLE

11/11

browser journey against
production — book, confirm,
cancel, slot returns

19

unit tests, run in CI against a
real Postgres 17

0

npm audit vulnerabilities

```
npm run test          # concurrency + lottery
npm run invariant     # whole-table sweep
npm run e2e           # 11 browser checks
```

The overlap count is never a check of the slot just tested — it is **every confirmed row against every other row** on the same court, run after every race.

STATED PLAINLY

WHAT IS DELIBERATELY OUT OF SCOPE

No passwords. Identity is a signed cookie over a seeded roster, with a “switch reader” control — so a judge can race two students without two browser profiles. Swapping in SSO replaces one function.

The demo endpoints are unauthenticated and rate limited rather than gated, because the race demo has to stay one click away.

What is enforced: ownership scoped inside every UPDATE, zod on every request body, bind parameters on every query, no `sql.unsafe` anywhere.

Next: institute SSO, push notifications for waitlist offers, and a real limiter keyed on identity rather than on a header the client controls.

WHAT THE PRODUCT ACTUALLY DOES

FEATURE	FUNCTIONALITY
Availability view	Twelve campus facilities with live open, taken and closed counts. Availability is derived from the bookings table per request — never stored, so it cannot drift.
Slot selection	A day grid per facility on its own timetable, in IST, seven days ahead. Updates live over Server-Sent Events.
Booking with a guarantee	Confirmed by the database or refused by it. Idempotency keys make a retry return the original booking rather than a second one.
Typed rejections	SLOT_TAKEN, QUOTA_EXCEEDED, OVERLAPS_OWN — each carrying three alternative slots and a queue position.
Waitlist	Join a full slot; a cancellation offers it to the next student in the <i>same transaction</i> , with a claim expiry.
Fair draw	Peak slots go to a seeded, weighted, published lottery, so the fastest wifi does not always win.
Operations console	Maintenance closures, which the same constraint refuses if a student already holds that window.
Insights	Demand by hour, a utilisation heatmap, no-show rate, and under-used prime slots.

FROM DISCOVERY TO CONFIRMATION

01 Browse Facilities with live open counts. Filter by sport, pick a day.	02 Inspect The day grid for one court. Open, taken, yours, closed — by fill, not hue.	03 Select Tap a slot. The sheet shows party size and the idempotency key.	04 Submit One constrained INSERT. No availability check on the write path.	05 Decide Postgres confirms exactly one and refuses every other request.
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WON → 201 CONFIRMED

A booking code is issued. It appears in My Bookings and the grid marks the slot as yours, live on every open screen. Cancelling releases it *and* promotes the waitlist in one transaction.

LOST → 409 TYPED REJECTION

SQLSTATE 23P01 becomes a reason a student can act on: three alternative slots on the same grid, plus a waitlist place that is offered automatically when somebody cancels.

WHAT IT IS BUILT ON

APPLICATION

Next.js 15	App Router, React Server Components
React 19	UI runtime
TypeScript	Strict mode
Tailwind CSS v4	Token layer, no component library
lucide-react	Icon set
zod	Validation on every request body
date-fns	IST-correct date handling

DATA AND PLATFORM

PostgreSQL 17	Neon, ap-southeast-1
btree_gist	Extension backing the exclusion constraint
postgres.js	Driver — tagged templates, bind parameters only
Vercel	Serverless functions, region sin1
Server-Sent Events	Live availability, native — no socket library
Vitest	19 unit and concurrency tests
Puppeteer	11-check browser journey
GitHub Actions	CI against a real Postgres service container

ONE PLATFORM. ZERO CLASHES.

FIFTY STUDENTS. ONE COURT. ONE WINNER.

Run the race yourself — the burst is fired at a real Postgres table through a real HTTP endpoint, and the whole table is swept afterwards.

innovait-hackathon.vercel.app/race

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